

Week 3 Development Documentation

NEXTERN

A Mobile Platform for Learning, Internships, and Career Development

Prepared by

TEAM 18

Mobile App Development with Flutter Internship

@Excelerate

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Week 3

Members

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1. Cross-App Functional Expansion and Workflow Improvements

The application was expanded beyond static screens to support more complete learner and administrator workflows.

Learner Experience

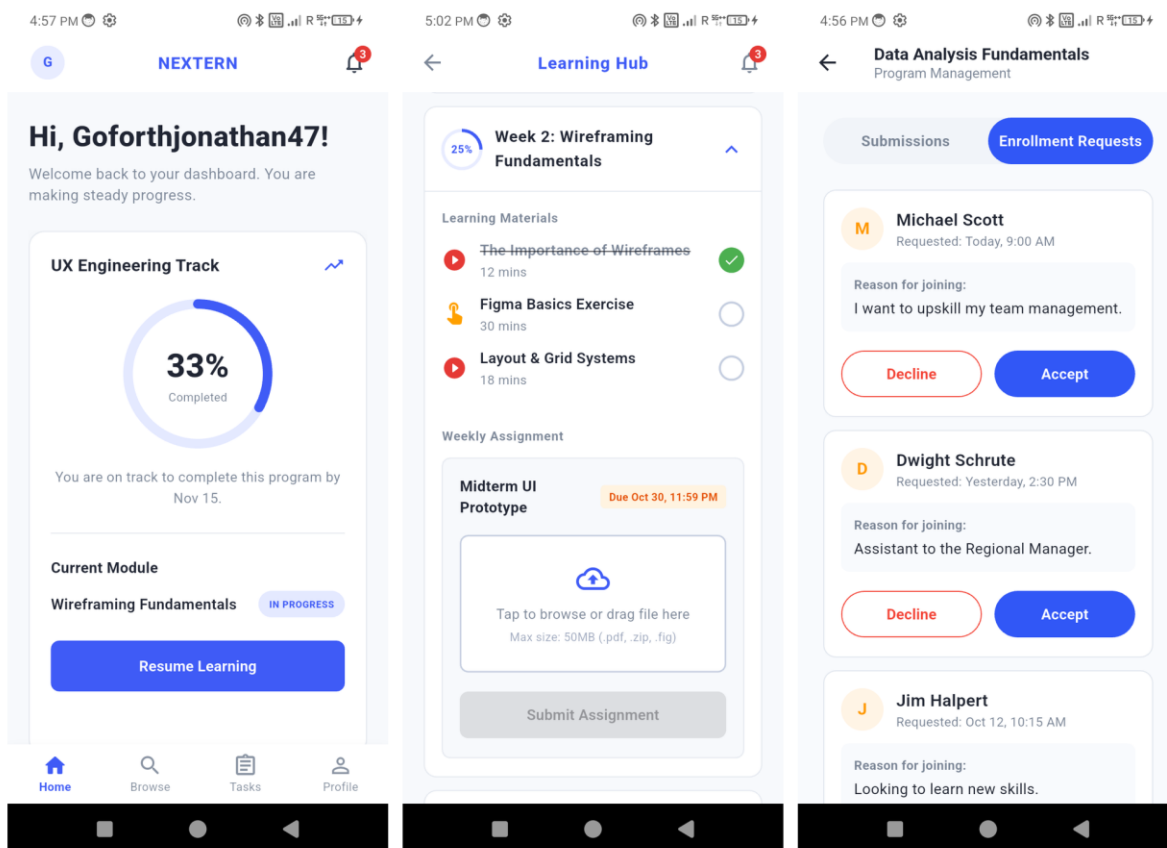
- The Home screen now includes a carousel for active programs, announcements, pending assignments with direct links to submission pages, saved programs, and functional notifications. Notifications support badges, dropdown views, mark-as-read, delete, and detail actions.
- The Browse screen initially shows three programs and includes a Load More option. Live search, quick filters, and a bottom-sheet filter make it easier to find programs by category or tag.
- The enrolment flow now supports visible application states such as Under Review, Accepted, and Rejected. Accepted learners can access a dedicated Program Learning screen with weekly collapsible modules, progress tracking, assignment drop boxes and working media or document viewers. Resume actions on the dashboard link directly back to active learning content.
- The Tasks area was divided into active coursework and downloadable certificates. A complete Learner Profile area was also added, with screens for editing profile information, notification settings, security, feedback, and Help Centre support.

Administrator Experience

- The Admin dashboard now shows total learners, active programs, and recent activity. The announcement creator was expanded to include a title, description, and resource attachments, with announcements synchronized to learner dashboards.
- The previous monitoring screen was replaced with a Program Manager system. Administrators can create, edit, and delete programs, add modules, set dates and submission deadlines, filter submissions, review learner work, open attached files, and accept or reject enrolment applications.
- A dedicated Users screen was added for searching, filtering, and managing learners. Available actions include viewing profiles, resetting passwords, and suspending accounts.
- The Admin Profile area now includes Edit Profile, Admin Preferences, Security, two-factor authentication, improved navigation, and correct logout routing.

Cross-App Refinements

- Learner and administrator notifications were separated by role, while announcements remain synchronized across both sides of the application. Login screens were also improved by repositioning Forgot Password and replacing text placeholders with official Google icons.
- A Figma application-flow diagram was created to show the main learner and administrator navigation paths using screenshots and directional arrows.



2. Dynamic Program Data and Shared Saved-Program State

Hard-coded program listings were replaced with a local JSON file that acts as a mock API. A reusable Program model was created in `lib/models/program.dart` to validate required fields and provide safe fallback values for optional data. The original five programs were moved to `assets/data/programs.json` and registered as Flutter assets.

Services and State Management

- ProgramService loads and parses the JSON file, removes inactive records, sorts programs by display order, and returns descriptive errors when the data is invalid.
- SavedProgramService uses SharedPreferences to store bookmarked program IDs so saved programs remain available after restarting the application.
- A singleton ProgramStore built with ChangeNotifier manages:
 - Program data
 - Loading state
 - Error state
 - Saved program IDs
- Its main methods include `initialize()`, `retry()`, and `toggleSaved()`.

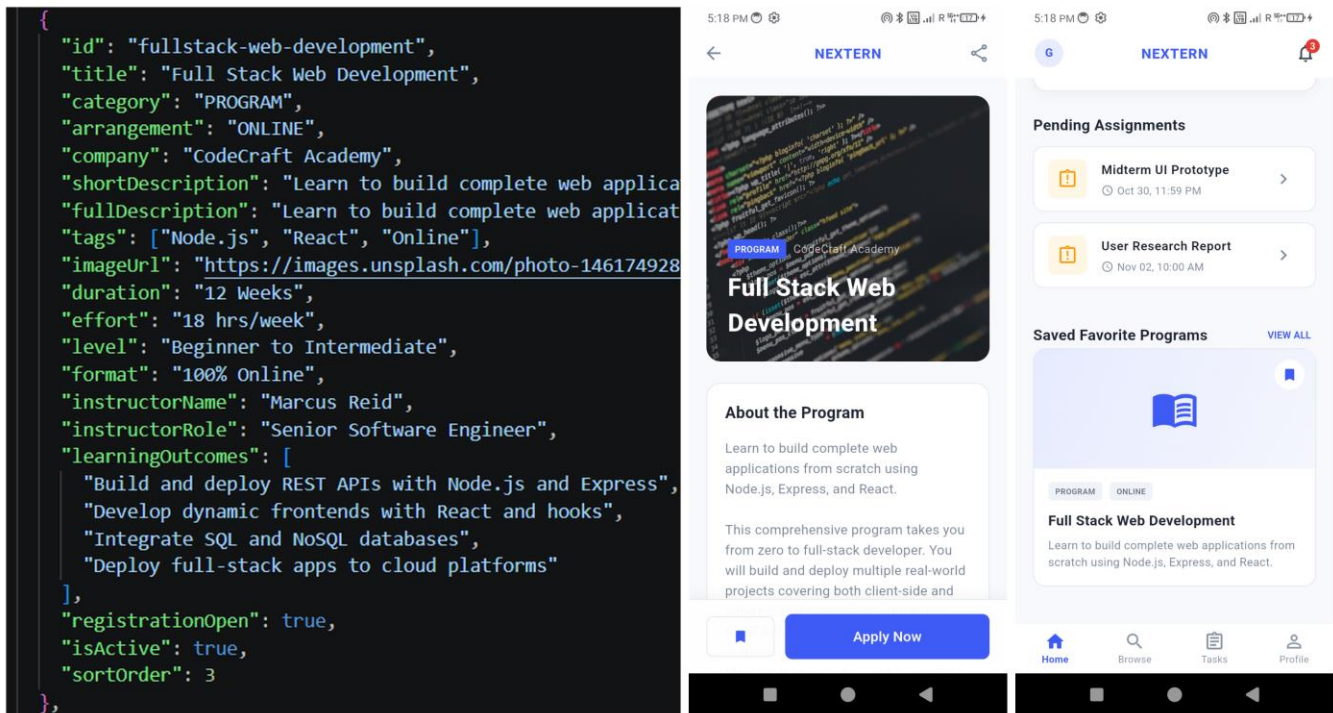
UI Integration

The Browse screen now reads from ProgramStore and includes loading, empty, error, and retry states. Search, filters, pagination, and bookmarking continue to work with the new data layer.

The Home screen's saved-program section now shows the learner's actual bookmarks instead of a fixed example.

The Program Details screen displays all information from the selected Program object, including image, description, duration, effort, level, format, instructor, learning outcomes, and saved status. Missing optional data is handled with fallback UI, such as placeholder images or hidden sections.

This implementation keeps Browse, Home, and Program Details synchronized while preparing the app for a future remote API.



3. Program Application Form and Enrolment Submission

- A reusable ProgramApplicationForm widget was created for program enrolment and connected to EnrollmentService.
- The form collects:
 - Experience level
 - Reason for joining
 - Optional portfolio URL
 - Confirmation checkbox
- Validation ensures that an experience level is selected, the reason field is completed, the optional portfolio URL is valid when entered, and the confirmation checkbox is selected.
- The form includes loading, success, and error states. During submission, repeated requests are blocked. Successful submissions update the learner's enrollment status, while failed submissions display an error and allow another attempt.
- The Program Details screen now opens the application form and reflects the latest enrollment state after submission. Applications can then move through Under Review, Accepted, or Rejected, with the same status shared between learner and administrator workflows.

The image displays three sequential screenshots of the NEXTERN mobile application interface, illustrating the application form and submission process.

Screenshot 1 (Left): Shows the 'Apply for Frontend Developer Intern - React/Tailwind' form. It includes fields for 'Experience Level' (a dropdown menu), 'Reason for joining' (a text area), and 'Portfolio URL (optional)' (a text field). Below these fields is a checkbox labeled 'I confirm that the information provided is accurate.' and a 'Submit Application' button.

Screenshot 2 (Middle): Shows the 'Apply for UX/UI Foundations Bootcamp' form. It includes fields for 'Experience Level' (a dropdown menu with 'Beginner' selected), 'Reason for joining' (a text area with 'It piqued my curiosity.'), and 'Portfolio URL (optional)' (a text field with 'om/in/jonathan-goforth-51637436a/'). Below these fields is a checkbox labeled 'I confirm that the information provided is accurate.' and a 'Submit Application' button. A red error message 'Please confirm the information before submitting.' is visible below the confirmation checkbox.

Screenshot 3 (Right): Shows the 'About the Program' screen for 'UX/UI Foundations Bootcamp'. It includes a 'PROGRAM' badge, a 'Nextern Academy' logo, and a description of the program. A white modal box with a checkmark icon and the text 'Application Submitted' and 'Status: Under Review' is overlaid on the screen.

4. Overall Result

The Week 3 update moved Nextern from a mainly visual prototype toward a functional Flutter application.

The project now includes dynamic JSON-backed program data, persistent saved programs, synchronized shared state, validated enrollment forms, application-status tracking, learner content, assignment submission tools, administrator review features, improved profiles, notifications, and clearer navigation.

The codebase is also more maintainable because models, services, state management, and interface components are separated. This creates a stronger foundation for connecting the application to a live backend in a future phase.

Project Links

1. **GitHub:** <https://github.com/Jonathan-Go4th/Project-Nextern>
2. **Demo Video:** <https://drive.google.com/file/d/1h6qsRaYFGqBf4dTSoSTcl3lorAMZbxYi/view?usp=sharing>
3. **Figma:** <https://www.figma.com/design/zFOUI2k38kZSWLhMXAM1sV/Nextern-App?node-id=92-55&t=ox2aUBD1k48iS4Lf-0>